

AFRICA MEDICAL COLLEGE

Final Examination in Histology and Embryology for Year I Medical Students

Time allowed: 3:00 hours.

Name: _____

Date: Feb 18, 2013

I. HISTOLOGY

For each of the following **Multiple Choice Questions** choose the one **BEST** appropriate answer and mark 'X' in the appropriate box located against the corresponding question number and letter in the **Separate Answer Sheet** provided.

_____1. 1000 micrometers is equivalent to:

- a) 1 centimeter. b) 100 millimeters. c) 10 millimeters. d) 1 millimeter. e) 0.1 millimeter

_____2. Which of the following organelles is the site of **steroid hormone synthesis?**

- a) Golgi complex b) mitochondria c) lysosomes d) RER e) SER

_____3. With regard to the ***cellular organelles***, which statement is ***true?***

- a) The nucleolus is the site of assembly of ribosomal subunits.
b) Lysosomal enzymes have optimal activity at basic pH to synthesize rRNA.
c) Golgi apparatus possess heterochromatin which is transcriptionally active.
d) The nucleolus is a well-defined, membrane-bounded acidophilic nuclear inclusion.
e) Phospholipids are synthesized by peroxisomes.

_____4. Concerning the **morphology of a cell**, select the **correct statement**.

- a) Microtubules are more stable than microfilaments.
b) G₀ phase is the longest and most variable cell-cycle phase in continuously dividing cells.
c) Ribosomes are membrane-bounded organelles and are involved in the breakdown of hydrogen peroxide.
d) Ion-transporting cells contain abundant basal mitochondria between infoldings of the basal plasma membrane.
e) Centrioles are associated with membrane activities such as endocytosis, exocytosis, and cell migratory activities.

_____5. With regard to the **epithelial tissue**, select the **incorrect statement**.

- a) The lining of the bladder is transitional epithelium.
b) Holocrine secretion is the typical mode of secretion of DNES cells.
c) An example of epithelioid tissue is the parenchyma of the adrenal gland.
d) Endothelium is simple squamous epithelium that lines the vascular system.
e) Merocrine secretion is characterized by discharge of a secretion product with no loss of cytoplasm or membrane.

———6. The stain of choice to demonstrate **glycogen and/or mucus** in cells is:

- a. periodic acid Schiff (PAS). b. hematoxylin. c. eosin. d. lipofuchsin. e. hemosiderin.

———7. Which **epithelial cell surface specialization** is used to move mucous and particles along the surface?

- a. cilia b. stereocilia c. desmosomes d. microvilli e. villi

———8. All **epithelia** rest on a thin layer called:

- a. submucosa. b. lamina propria. c. dermis. d. basement membrane [basal lamina]. e. serosa.

———9. Which surface is **NOT lined by epithelial tissue**?

- a. surface of skin. b. cornea surface of eye. c. luminal surface of intestinal tract. d. luminal surface of trachea and alveoli of lungs. e. surfaces of bones and muscles.

———10. With regard to the **epithelial tissue**, which statement is **true**?

- a) Arteriovenous anastomosis provides nutrition to the highly cohesive rows of cells.
b) Desmosomes permit the passage of ions between adjacent cells.
c) Microvilli contain sheaf of microtubules with a 9+2 axoneme pattern.
d) The major constituents of epithelia is abundant extracellular matrix with amorphous ground substance.
e) In pseudostratified columnar epithelium all cells touch the basal lamina.

———12. One of the following structures is lined by **stratified columnar epithelium**.

- a) Urinary bladder. b) Conjunctiva. c) Esophagus. d) Vagina. e) Trachea.

———13. **Mesothelium** lines:

- a. the respiratory and the urinary tracts. b. the peritoneal, pleural and pericardial cavities.
c. the nasal and the oral cavities. d. the gastrointestinal and reproductive tracts
e. the central nervous system and the muscular system

———14. With regard to the **connective tissue**, which statement is **false**?

- a) White adipose tissue tends to increase as humans age.
b) Mucous connective tissue lacks connective tissue cells.
c) In ordinary connective tissue, the predominant component of ground substance is water.
d) Fibroblast produces and maintains all extracellular matrix components of connective tissue.
e) Plasma cells, which are derived from mast cells, characteristically synthesize and secrete immunoglobulins.

———15. Which **connective tissue cell type** is functionally related to the **basophils**?

- a) Fibroblast b) Myofibroblast c) Histiocyte d) Plasma cell e) Mast cell

———16. Which **connective tissue component** is characterized by high tensile strength and flexibility?

- a. collagen b. elastin c. glycosaminoglycan d. fibroblasts e. lipid

———17. Cells which secrete the components of **bone matrix** are called:

- a. fibroblasts. b. chondroblasts. c. osteoblasts. d. osteoclasts. e. osteocytes.

———18. With regard to **cartilage**, which statement is **false**?

- a) Hyaline cartilage is characterized by an amorphous glassy matrix.

- b) Fibrocartilage has thick bundles of collagen fibres in the matrix.
- c) Fibrocartilage is present in regions subjected to high tension.
- d) Elastic cartilage is similar to hyaline cartilage except the abundant elastic fibres it contains.
- e) Fibrocartilage is used for endochondral bone ossification.

—————19. **Bone formation** without a pre-existing cartilage model is called:

- a. intramembranous bone formation.
- b. extramembraneous bone formation.
- c. endochondral bone formation.
- d. trabecular bone formation.
- e. cancellous bone formation.

—————20. In the **restructuring of the skeleton, bone remodeling**:

- a. follows a cycle in which bone formation alternates with resorption.
- b. occurs exclusively in adult bone.
- c. occurs independently of soft tissue growth and external forces.
- d. only occurs at the epiphyses of long bones.
- e. results in the formation of cement lines.

—————21. **Elastic cartilage,**

- a) is found in the annulus fibrosus and intervertebral disks.
- b) is the primary skeletal tissue in the fetus.
- c) has no identifiable perichondrium.
- d) the collagen in its matrix is mainly type II.
- e) is the most widely distributed cartilage type in the body.

—————22. Which of the following cells **does not** belong to the **mononucleated phagocytic system**?

- a) Merckel's cells
- b) Dust cells
- c) Langerhan's cells
- d) Kupffer cells
- e) Cells of Ito

—————23. Which of the following statements is **correct** regarding the **bone**?

- a) Primary bone has a higher mineral content than the secondary bone.
- b) Sharpey's fibers are calcified collagen bundles that run in concentric manner.
- c) Osteoclasts have processes that run in their canaliculi.
- d) Osteoblasts derived from blood monocytes.
- e) Bone contains matrix with more inorganic components than organic component in its dry weight.

—————24. **Compact (dense) bone,**

- a) is the predominant bone tissue in the epiphyses of adult long bones.
- b) is characterized by the presence of osteons.
- c) is also called cancellous bone.
- d) can be either primary or secondary bone.
- e) is typically more radiolucent than other types of bones.

—————25. With regard to the **muscular system**, which statement is **false**?

- a) The distance between two Z-lines in skeletal muscle is equivalent to the length of the sarcomere.
- b) Z-lines represent attachment sites for thick filaments (myosin).
- c) The electrical signal of an action potential is conveyed into the interior of each muscle fiber by transverse tubules.
- d) Cardiac muscle consists of individual cells attached end-to-end by intercalated disks.
- e) Skeletal muscle can be described as syncytial, with many nuclei per "cell" located at the periphery of its muscle fibers, close to the fiber membrane.

—————26. The sense organ most closely associated with **stretch reflexes** (e.g., knee jerk) is the:

- a. Meissner's corpuscle.
- b. Pacinian corpuscle.
- c. muscle spindle.
- d. Golgi tendon organ.
- e. free nerve ending.

—————27. Which of the following cell types is found predominantly in **white matter**?

- a) fibrous astrocytes
- b) microglia
- c) satellite cells
- d) protoplasmic astrocytes
- e) Schwann cells

———28. With regard to the **dendrites**,

- a) each has a constant diameter along its entire length.
- b) contain neurosecretory vesicles.
- c) may contain Nissl bodies.
- d) are found in white matter.
- e) are typically myelinated.

———29. Which of the following statements is **true** concerning the **nervous tissue**?

- a) Schwann cells provide myelin for several axons simultaneously.
- b) Unmyelinated axons generally have a smaller diameter than myelinated axons.
- c) Action potentials normally travel from the axon terminal toward the cell body.
- d) Gray matter in the central nervous system consists of axon tracts with abundant myelin.
- e) Microglial line the central canal of the spinal cord and the ventricular system of the brain.

———30. A typical **peripheral mixed nerve** includes all of the following **EXCEPT**:

- a) Schwann cells.
- b) astrocytes.
- c) sensory axons
- d) motor axons
- e) connective tissue of the perineurium and/or endoneurium

———31. **Dura mater** consists of:

- a. loose connective tissue with CSF as ground substance.
- b. dense fibrous connective tissue.
- c. white matter.
- d. gray matter.
- e. a thin layer of cells and collagen adhering to outer surface of the brain.

———32. Which of the following statements is **false** about the **cerebrospinal fluid**?

- b) Arachnoid mater forms trabeculae through which cerebrospinal fluid flows.
- b) Compared to blood, normal cerebrospinal fluid contains much higher levels of potassium and calcium.
- c) Cerebrospinal fluid is actively secreted by ependymal cells of the choroid plexus.
- d) An imbalance between production and drainage of cerebrospinal fluid can cause hydrocephalus.
- e) None of the above.

———33. Which of the following statements is **true** regarding the **nervous system**?

- a) Purkinje cells are characteristic of the cerebral cortex.
- b) White matter consists mainly of axons and neuroglial cells.
- c) Choroid plexus is found in the central canal of the spinal cord.
- d) Pyramidal cells are characteristic of the ventral horn of spinal cord.
- e) Ependymal cells help regulate the interstitial environment of central nervous tissue, serving as a buffer between nerve cells and capillaries.

———34. Which of the following statements is **not true** regarding the **supporting cells in the nervous tissue**:

- a) Microglia are the smallest in size among the neuroglial cells.
- b) Astrocytes serve in conducting metabolites between blood vessels and neurones.
- c) Satellite cells surround the perikarya of the spinal ganglia in a stratified cuboidal fashion.
- d) Oligodendrocytes are found in both gray and white matters of the central nervous system.
- e) Ependymal cells line the central canal of the spinal cord and the ventricular system of the brain.

———35. Which of the following tissue types **does not** include **secretion** as a major function of its constituent cells?

- a) connective tissue
- b) epithelial tissue
- c) muscle tissue
- d) nerve tissue
- e) all of them

———36. Which of the following statements is **false** with regard to the **circulatory system**?

- a) **Lymphatic capillaries can penetrate the media of veins.**
- b) **Vasa vasorum are more frequent in veins than in arteries.**

- c) The tunica adventitia tends to be much larger in veins than in arteries.
- d) The concentration of glycosaminoglycans is higher in arterial than in venous tissue.
- e) Sinusoids of the liver and the glomeruli of the kidney typically possess continuous endothelium.

—————37 Tunica intima, supported by a thin layer of connective tissue:

- a) contains some small capillaries.
- b) is homologous to the pericardium of the heart.
- c) includes a layer of simple squamous epithelium.
- d) is a fenestrated sheet of elastin.
- e) is separated from tunica media of arteries by the external elastic lamina.

—————38. Which is (normally) the **most common white blood in circulating blood?**

- a. lymphocyte.
- b. monocyte.
- c. neutrophil.
- d. eosinophil.
- e. basophil.

—————39. Which of the following blood cell types will normally **re-enter the bloodstream after leaving circulation and immigrating into peripheral tissues?**

- a. lymphocyte.
- b. monocyte.
- c. neutrophil.
- d. eosinophil.
- e. basophil.

—————40. A **macrophage** can originate from a:

- a. lymphocyte.
- b. monocyte.
- c. neutrophil.
- d. eosinophil.
- e. basophil.

—————41. *A prominent inner elastic membrane (internal elastic lamina), often appearing in cross section*

asa wavy or sinuous line, is characteristic of:

- a. muscular arteries.
- b. small veins.
- c. large veins.
- d. venules.
- e. capillaries.

—————42. *Continuous endothelium is found in:*

- a. endocrine organs.
- b. kidneys.
- c. lungs.
- d. spleen.
- e. intestinal villi.

—————43. *Which of the following statements is not correct regarding the veins?*

- a) Are more in number than the arteries
- b) Have prominent internal elastic lamina.
- c) Have relatively well developed connective tissue and less developed muscles when compared with the arteries.
- d) Have thinner wall and larger lumen when compared with the arteries.
- e) Most veins have no clear boundary between tunica intima and media.

—————44. *With regard to impulse generating and conducting system of the heart:*

- a) Sinuatrial node is located on the lower aspect of the interatrial septal wall of the left atrium.
- b) Cells of the sinuatrial node are larger in size when compared to those of the ordinary cardiac myocytes.
- c) Cells of the sinuatrial node have more myofibrils than those of the ordinary cardiac myocytes.
- d) Atrioventricular node is bigger in size than the sinuatrial node.
- e) Purkinje cells, when seen under light microscopy following H & E staining, appear as large cells with clear zone surrounding centrally placed nucleus and denser peripheral cytoplasm.

—————45. *Which of the following statements is false with regard to the circulatory system?*

- a. Post capillary venules lack a smooth muscle media.
- b. Carotid bodies are chemoreceptors sensitive to hypoxia.

c. The brachiocephalic and subclavian arteries are elastic arteries.

d. Continuous capillaries often have phagocytic cells inserted between the endothelial cells of their lining.

e. Venous valves of the lower limbs have two semilunar folds that are lined on both sides by endothelium.

———46. Smooth muscle is most substantial in:

a. the epicardium of the heart.

b. the media of elastic arteries.

c. the media of muscular arteries.

d. the myocardium of the heart.

e. arterioles.

———47. The sino-atrial (SA) node, the atrio-ventricular (AV) node, and the Purkinje fibers of the myocardium all consist of specialized:

a. endothelial cells.

b. fibroblasts.

c. nerve cells.

d. smooth muscle cells.

e. cardiac muscle cells.

———48. Pericytes are

a) specialized cardiac muscle cells

b) are attached to the outside of capillaries

c) specialized smooth muscle cells

d) terminally differentiated

e) multinucleated

———49. A differential cell count of a blood smear from a patient with a parasitic infection is likely to reveal

an increase in the circulating numbers of which of the following blood cell types?

a) basophils

b) eosinophils

c) neutrophils

d) monocytes

e) platelets

———50. Smooth muscle is most substantial in:

a. the epicardium of the heart.

b. the media of elastic arteries.

e. arterioles.

c. the media of muscular arteries.

d. the myocardium of the heart.

———51. Heart valves normally consist of an endothelial surface covering:

a. cardiac muscle fibers.

b. hyaline cartilage.

c. loose areolar connective

tissue. d. fibrocollagenous and fibroelastic connective tissue.

e. adipose connective

tissue.

———52. Which of the following statements is false with regard to the histology of the heart?

a. Ventricles contain more abundant elastic fibers than do the atria.

b. Adipose tissue that generally surrounds the heart accumulates in the epicardium.

c. The coronary arteries branch to supply the epicardium, myocardium, and endocardium,

d. Endocardium is the thickest in atria and the thinnest in ventricles, especially the left ventricle.

e. Cardiac muscle is more vascularized and has more abundant mitochondria than does skeletal muscle.

———53. Which of the following substances is present in higher concentrations in plasma than in serum?

a) albumin

b) fibrinogen

c) immunoglobulin

d) glucose

e) serotonin

———54. Which of the following immunoglobulin types is most important in conferring passive immunity on the fetus and on the newborn infant?

a) IgA

b) IgD

c) IgE

d) IgG

e) IgM

———55. Which of the following is partly **encapsulated and covered by nonkeratinized stratified squamous epithelium?**

- a) appendix b) lingual tonsil c) palatine tonsil d) Peyer's patch e) pharyngeal tonsil

———56. Which of the following **lymphoid organs** receives **afferent lymphatic vessels?**

- a) appendix b) palatine tonsil c) spleen d) thymus e) lymph node

———57. **Thymus-dependent** region of a peripheral lymphoid organs include,

- a) medullary cords of lymph nodes. b) Hassall's corpuscles of the thymus.
c) crypts of the pharyngeal tonsils. d) periaarterial lymphatic sheaths of the spleen.
e) red pulp cords (Billroth's cords) of the spleen.

———58. The **thymus:**

- a) Has blood-thymus barrier in the cortex. b) Is relatively more developed with increasing age.
c) Among blood vessels, contains only sinusoids. d) Has lymphoid nodules in its cortex.
e) Contains more B lymphocytes than the T lymphocytes.

———59. The **spleen:**

- a) Has smooth myocytes in its capsule. b) Filters lymph. c) Has no B lymphocytes.
d) Has an outer cortex and inner medulla. e) Contains periaarterial lymphoid sheath in its red pulp.

———60. Which of the following components of the **respiratory tract wall increase(s) in amount from trachea to alveoli?**

- a) cilia b) elastic fibers c) smooth muscle d) cartilage e) goblet cells

———61. Which of the following structures is (are) partly lined by **nonkeratinized stratified squamous epithelium?**

- a) nasopharynx b) larynx c) bronchi d) trachea e) bronchioles

———62. In which of the following are goblet cells most frequent?

- a. trachea b. bronchus c. bronchiole d. alveoluse.

The frequency of goblet cells in each of the above is about equal.

———63. Nasal mucosa includes:

- a. pseudostratified, ciliated columnar epithelium. b. lamina propria. c. seromucous glands.
d. venous plexuses. e. all of the above.

———64. As respiratory passages branch from trachea to bronchioles, the epithelium gradually changes from:

- a. simple squamous to ciliated, pseudostratified columnar. b. simple columnar to simple squamous.
c. simple cuboidal to simple columnar. d. stratified squamous to simple squamous. e. ciliated, pseudostratified columnar to simple cuboidal.

———65. Alveolar type I cells are:

a. cuboidal cells that secrete surfactant.

c. ciliated cells that move mucous.
secretion.

e. amoeboid, phagocytic cells that clean the alveolar surface.

b. squamous cells involved in gas exchange.

d. columnar cells that are rich in serous

———66. **Bronchioles:**

a) Have well developed hyaline cartilage

c) Have no smooth muscle

b) Are lined by stratified squamous epithelium

d) Have Clara cells

e) Have mucous producing glands

———67. **Pulmonary surfactant,**

d) prevents alveolar collapse by increasing surface tension.

c) consists of a monolayer of phospholipids overlying an aqueous hypophase.

e) excessive production in the lungs of premature infants causes respiratory distress syndrome.

a) is secreted by type I pneumocytes.

bacteria.

b) may promote the growth of

———68. **The lining of the pleural cavity is:**

a. endothelium.

b. mesothelium.

c. pseudostratified ciliated columnar

epithelium.

d. type I pneumocytes.

e. type II pneumocytes.

———69. **Which of the following type of cells is not present in the respiratory epithelium of the nasal cavity?**

a) Ciliated columnar cells

b) Goblet cells

c) Basal cells

d) Stem cells

e) None of the above

———70. **Arrector pili muscles,**

a) are composed of bundles of myoepithelial cells b) help push sebum onto the skin surface.

c) have no effect on hair position in humans. d) contain capillary networks that nourish growing hairs.

e) attach directly to the external root sheath by desmosomes.

———71. **The glands most closely associated with hair follicles are:**

a. eccrine sweat glands.

b. apocrine sweat glands.

c. sebaceous glands. d.

lymphatic glands.

e. submucosal glands.

———72. **Which of the skin's many functions are partly or completely provided by components of the dermis?**

a) protection from dehydration

b) regulation of blood pressure

c) protection from ultraviolet radiation

d) synthesis of keratin e) synthesis of melanin

———73. **Which of the following epidermal layers is most likely to contain both desmosomes & hemidesmosomes?**

a) stratum basale

b) stratum corneum

c) stratum granulosum

d) stratum lucidum

e) stratum spinosum

———74. **Which of the following is (are) often absent in thick skin but typically present in thin skin?**

a) arrector pili muscles

b) Meissner's corpuscles

c) sweat glands

d) stratum basale

e) stratum spinosum

————75. **Cells of epidermis** that form the stratified squamous keratinized epithelium are:

- a) Keratinocytes b) Melanocytes c) Langerhan's cells d) Merckel's cells e) Chondrocyte

————76. **Sebaceous gland** is a type of:

- a) Compound tubular gland. b) Compound alveolar gland. c) Simple straight tubular gland.
d) Simple branched alveolar gland. e) Simple coiled tubular gland.

————77. Which of the following statements is **true** about the **lip**?

- a) The inner aspect has sebaceous glands. b) The outer aspect has apocrine sweat glands.
c) The transitional border has abundant salivary glands.
d) The inner surface is lined by stratified columnar epithelium. e) The core contains skeletal muscles.

————78. Which of the following statements is **true** regarding the **teeth**?

- a) Odontoblastic processes branch and taper as they approach dentinoenamel junction.
b) Dentin is not present in the deciduous teeth. c) Enamel has more collagen fibres than dentin.
d) Enamel is found overlying dentin in both crown and root portion of the teeth.
e) Enamel develops from mesoderm, while both dentin and cementum develop from overlying ectoderm.

————79. **Muscularis mucosae of esophagus**:

- a) Is made by inner oblique, middle circular and outer longitudinal layers.
b) Is made by inner circular and outer longitudinal layers. c) Is found only as longitudinal layer.
d) Is entirely of skeletal muscle type. e) None of the above.

————80. Which of the following statements is **true** about the **stomach**?

- a) Characteristically contains the most goblet cells in the mucosa covering its villi.
b) Has basophilic oxyphil cells in its compound branched tubuloalveolar glands that secrete HCl.
c) Mucosal glands of the pylorus contain abundant parietal cells lined with stratified squamous nonkeratinized epithelium in their basal half.
d) Has its muscularis externa made by skeletal muscle fibres. e) Mucosal glands of the pylorus secrete mainly mucus.

————81. Which of the following is **correct** about the **histology of pancreas**?

- a) The exocrine cells are arranged as simple tubular type.
b) The exocrine portions have more capillaries than the endocrine portion.
c) The islets of Langerhans are made by larger and more eosinophilic cells than the exocrine cells.
d) Although it is a mixed exocrine and endocrine gland, the majority of the secretory cells are exocrine cells.
e) The B cells of islets of Langerhans are distinguished under light microscope by being larger and more basophilic than the other cells of the islets.

————82. Which of the following statements is **correct** regarding the **intestine**?

- a) Intestinal glands in the large intestine are branched alveolar type.
b) The striated border of the small intestine is made by microvilli.
c) Small intestine has taenia coli. d) Large intestine has villi.
e) Both the small and the large intestines are able to withstand the acidic environment because they are lined by stratified squamous parakeratinized epithelium.

————83. Which one of the following marks best **ileum** from **duodenum** and **jejunum**?

- a) Long and finger like villi. b) Well developed Peyer's patches. c) Well developed plicae circulares.
d) Well developed microvilli. e) Simple columnar lining epithelium.

———84 **Permanent circular folds (plicae circulares)** is a characteristic structural feature of the walls of which of the following organs?

- a) anal canal b) colon c) cecum d) small intestine e) stomach

———85. In the liver, **Kupffer cells:**

- a) are nonphagocytic b) contain lysosomes c) are derived from circulating neutrophils
d) resemble hepatocytes in HE-stained sections e) border on the bile canaliculi

———86. Select the **correct** statement with regard to the **GI tract**.

- a) Brunner's glands empty their secretions into the lacteals.
b) Parotid gland lacks serous acini but contains mucous demilunes.
c) A hepatic cord is the tough connective tissue surrounding a portal triad.
d) The whitish tips of filiform papillae consist of keratinized epithelial tissue.
e) The myenteric (Auerbach's) plexus is located between the lamina propria and the muscularis mucosa.

———87. **Submucosal glands** occur in the:

- a. duodenum and esophagus. b. Stomach and ileum. c. jejunum and colon.
d. appendix and anal canal. e. All of them.

———88. Select the **incorrect** statement with regard to the **GI tract**.

- a. Folds in the mucosa of the stomach are called rugae.
b. Lymphatic vessels within intestinal villi, which transport lipids to the circulatory system, are called lacteals.
c. The basal surface of hepatocytes is bathed by blood plasma as it flows freely through the space of Disse.
d. The principal secretory product for cells of Brunner's (duodenal) glands is acid mucus.
e. The principal secretory product for the Paneth cells clustered at the ends of intestinal crypts is lysosomal enzymes.

———89. Which of the following regions of the **GI tract** is characterized by **stratified squamous epithelium**?

- a. duodenum b. jejunum c. ileum d. colon e. anal canal

———90. **Hepatocyte** function includes all of the following **EXCEPT**:

- a. drug detoxification. b. protein synthesis. c. gluconeogenesis.
d. urea formation by deamination of amino acids. e. breakdown of hemoglobin.

———91. Like the small intestine, the **gall bladder**:

- a. is lined by columnar cells with a brush border of microvilli. b. has goblet cells in the mucosal epithelium.
c. has crypts with Paneth cells at the bottom. d. has villi with lacteals in the core.
e. has a distinct muscularis mucosa, and is further surrounded by two distinct layers of circular and longitudinal muscle.

———92. With regard to the **digestive system**, which is the **correct statement**?

- a) Vermiform appendix is a blind-ended evagination of the ileum.
b) Myoepithelial cells are responsible for emptying the gallbladder.
c) Intrinsic factor activates the small intestine to stimulate calcium uptake in the digestive tract.
d) Filiform papillae lack taste buds. e) Odontoblasts synthesize enamel.

———93. Which of the following is **not** a function of the kidney?

a. Erythropoietin production
d. Aldosterone production

b. Vitamin D modification
e. Renin production

c. Acid base balance

———94. Select the **correct statement** pertaining the **morphology of the kidney**.

- a. Macula densa cells are located in the urinary pole of the nephrons.
- b. The distal convoluted tubules exhibit brush borders and apical canaliculi.
- c. The renal pyramid and its associated cortex is referred to as the renal lobe.
- d. Interstitial cells in the renal medulla secrete erythropoietin through filtration slits.
- e. Renin is secreted by the podocytes from the visceral layer of Bowman's capsule referred to as area cribrosa.

———95. In the **urinary system**, which statement is **false**?

- a. Aldosterone acts on the distal convoluted duct.
- b. Peritubular capillaries arise from the efferent arterioles.
- c. Endothelial cells of glomerular capillaries are of non-fenestrated variety.
- d. Epithelial cells lining the thin limb of the loop of Henle are simple squamous.
- e. The ureter is lined by transitional epithelium and it does not contain lymphatics.

———96. Which of the following is **not** the function of **mesangial cells**?

- a. synthesis of extracellular matrix.
- b. endocytosis and contractility.
- c. maintenance of acid-base balance in the blood.
- d. structural support to the glomerulus.
- e. increase blood flow and the effective surface area available for filtration.

———97. Which of the following hormones is carried to its target(s) by the **hypophyseal portal system**?

- a) ADH
- b) ACTH
- c) FSH
- d) TRH
- e) TSH

———98. Which statement is **true** with regard to the **endocrine system**?

- a) The chromophores of pars distalis of adenohypophysis maintain favorable fluid and electrolyte environment for the chromophil secretory cells.
- b) The secretory activity of the epiphysis cerebri {pineal body} is stimulated by darkness and inhibited by light.
- c) Calcitonin, secreted by the parathyroid gland, stimulates calcium resorption in the bones.
- d) The cells of the zona fasciculata of the suprarenal gland are called chromaffin cells because the granules of these cells can be stained with potassium bichromate.
- e) The basophilic oxyphil cells are the predominant cell type in the parathyroid gland which inhibit the activity of osteoclasts by decreasing blood calcium levels .

———99. The **neurohypophysis**,

- a) contains chromophils.
- b) includes the pars intermedia.
- c) secretes ACTH.
- d) contains axons of hypothalamic neurons.
- e) hormone production is regulated by releasing factors.

———100. **Testis** has the following coverings **except**:

- a. tunica adventitia
- b. tunica vasculosa
- c. tunica albuginea
- d. tunica vaginalis
- e. None

———101. In the **testis**, which statement is **true**?

- a. Interstitial cells (of Leydig) produce testicular fluid which constitute the bulk of ejaculate volume.
- b. The seminiferous tubules are lined with a complex stratified epithelium.
- c. The secondary spermatocytes are the largest and long-lived cells of the spermatogenetic lineage.
- d. The tunica vaginalis is thickened on the posterior surface to form the mediastinum and rete testis.
- e. During spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.

———102. Which of the following statements is **false** in the **male reproductive system**?

- a. The penile urethra passes through the corpus spongiosum.
- b. The secretion of seminal vesicle is particularly rich in fructose.
- c. The prostate gland is composed of compound tubuloalveolar glands.
- d. The ejaculatory duct connects the seminal vesicle with the prostatic urethra.
- e. Tunica albuginea, which is lined with pseudostratified columnar epithelium, is the thickest in epididymis.

———103. **Semen** contains fluid derived from the testes and all of the following **EXCEPT**:

- a. epididymis and vas deferens.
- b. seminal vesicles.
- c. bulbourethral glands.
- d. prostate.
- e. corpus spongiosum and corpora cavernosa.

———104. Which of the following statements is **true** in the **female reproductive system**?

- a. The proliferative phase of the endometrium is characterized by a stratified squamous surface.
- b. Germinal (surface) epithelium covering the ovary gives rise to ova.
- c. After ovulation, granulosa cells and cells of theca interna grow and differentiate to form the corpus luteum.
- d. Estrogen and progesterone are secreted by the gonadotropic cells of the adenohypophysis.
- e. The basic pH in the vaginal lumen is the secretion of HCl by the tubuloalveolar glands of the vaginal walls.

———105. Which of the following is true of the **ampulla** of the **uterine tube** but **not** of its **isthmus**?

- a) Contains large mucosal folds that are extensively branched.
- b) Contains smooth muscle in its walls.
- c) Is covered by serosa derived from the peritoneum.
- d) Contains ciliated epithelial cells.
- e) Contains mucus-secreting peg cells.

———106. Which of the following statements is **true** about the **female reproductive system**?

- a. Mucosa of the uterus is commonly called perimetrium.
- b. Fornix of vagina is lined by stratified squamous epithelium.
- c. The life span of corpus luteum of pregnancy is about 10 lunar months.
- d. Primary oocyte completes its first meiotic division at the time of fertilization.
- e. Corpora albicans are ovarian follicles which degenerate after failing to mature.

———107. Which of the following statements is **false** about the **sense organs**?

- a. There are more pain receptors than cold receptors in the body.
- b. Fingertips have more touch receptors than the skin of the back.
- c. End bulbs of Krause possess receptors for cold and pressure reception.
- d. Pacinian corpuscles are large, ellipsoid encapsulated, rapidly-adapting receptors.
- e. Meissner's corpuscles are unencapsulated, myelinated receptor terminals located in the deep dermis and hypodermis.

———108. The **cornea**,

- a) constitutes the anterior half of the tunica fibrosa.
- b) is composed of 10 layers.
- c) is avascular.
- d) accounts for eye color.
- e) is attached to the ciliary body by zonules.

———109. With regard to the **eye**, one of the following statements is **true**.

- a) Aqueous humor is actively secreted by cells of the stroma.
- b) Inner nuclear layer of the retina contains nuclei of photoreceptor cells.
- c) Lacrimal canaliculi are lined by a stratified ciliated columnar epithelium.
- d) An imbalance between production and drainage of aqueous humor can cause glaucoma.
- e) The cornea is covered by an incomplete layer of pigmented cells and fibroblasts on its anterior surface.

—————110. In the **ear**, one of the following statements is **false**.

- a) Otoliths are normally found in the saccule and utricle.
- b) endolymphatic sac contains phagocytic cells like macrophages and neutrophils.
- c) the scala vestibule and tympani communicate with each other at the helicotrema.
- d) the structure of the internal ear that contains special auditory receptors is the organ of Corti.
- e) tectorial membrane is secreted by the striavascularis of the ampullae of the semicircular canals.

—————111. With regard to the **ear**, one of the following statements is **true**.

- a) the bony labyrinth is filled with endolymph.
- b) The round window separates the middle ear from the eustachian tube.
- c) Reissner's membrane separates the scala media from the oval window.
- d) The middle ear contains modiolus, cristae ampullaris and semicircular canals.
- e) Hair cells of the ear, which are columnar mechanoreceptor cells, are bipolar neurons of the spiral ganglion.

—————112. Which type of **neurones of the cerebral cortex** are predominantly found in the **primary sensory cortex**?

- a) Stellate cells
- b) Horizontal cells of Cajal
- c) Fusiform cells
- d) Cells of Martinotti
- e) Pyramidal cells

—————113. The correct order of the three **layers of cerebellar cortex**, outer to inner is:

- a) Molecular layer, Purkinje cell layer and granular layer.
- b) Molecular layer, granular layer and Purkinje cell layer.
- c) Purkinje cell layer, molecular layer, and granular layer.
- d) Granular layer, molecular layer, Purkinje cell layer.
- e) Granular layer, Purkinje cell layer and molecular layer.

—————114. Which of the following statements is **not true** regarding the **meninges**?

- a) Cranial dura mater has an outer layer which is connected by Sharpey's fibres to the calvaria.
- b) Dura mater of the spinal cord is made up of only one layer of dense fibrous membrane.
- c) Arachnoid mater is delicate and transparent impermeable membrane.
- d) Arachnoid mater has numerous trabeculae which are connected to the dura mater traversing the subdural space.
- e) Arachnoid granulations are sites of formation of cerebrospinal fluid into the ventricle.

—————115. Which of the following is involved in the formation of **blood brain barrier**?

- a) Astrocytes
- b) Ependymal cells
- c) Oligodendrocytes
- d) Microglia
- e) Schwann cells

I. EMBRYOLOGY

1. Concerning **gametogenesis**, which statement is **true**?

- a) All oogonia become primary oocytes during ovulation.
- b) The difference in sex chromosome complement of oocytes forms the basis of primary sex determination.
- c) Secondary oocyte complete its second meiotic division to become a mature ovum at fertilization.
- d) Primary oocytes undergo cytodifferentiation to form three polar bodies prior to ovulation.
- e) Primary spermatocytes begin the first meiotic division before birth, but completion of prophase does not occur until adolescence.

2. Which statement is **true** about **female reproductive system**?

- a) the usual site of fertilization is the ampulla of the uterine tube.
- b) if an oocyte is fertilized, the corpus luteum involutes and degenerates.
- c) the endometrium has a wall made up of a very thick smooth muscle layer.
- d) the perimetrium is a trumpet-shaped opening towards the peritoneal cavity.
- e) the vaginal glands are sloughed off and discarded with each menstrual flow during pregnancy.

3. With respect to the **zonapellucida**, which statement is **false**?

- a. is composed of sulfated glycoproteins.
- b. prevents polyspermy
- c. prevents premature implantation
- d. functions as a porous filter
- e. arises by delamination from the embryoblast.

4. Before a **spermatozoan fertilizes an ovum**:

- a) its glycoprotein coat must be removed from its head.
- b) the zonapellucida must be shed.
- c) its acrosomal membrane must be impermeable.
- d) its tail piece should be lost.

e) it should be capacitated in the epididymidis.

5. With regard to fertilization, which statement is false? Fertilization:

- a) results in variation of human species through mingling of maternal and paternal chromosomes.
- b) causes metabolic activation of the ovum and initiates cleavage of the zygote.
- c) restores the normal diploid number of chromosomes in the zygote.
- d) determines chromosomal sex of the embryo.
- e) None of the above.

6. During the first week of development,

- a) The prochordal plate marks the site of the future mouth.
- b) The embryo is a solid ball of blastomeres and is called gastrula.
- c) The blastocyst is superficially implanted in the uterine wall (endometrium).
- d) Developmental disturbances may give rise to major congenital anomalies.
- e) The amniotic cavity is developed within the yolk sac as a space between the cytotrophoblast and the embryoblast.

7. Which of the following is/are not a derivative(s) of the neural crest?

- a) Schwann cells b) chromaffin cells c) neurohypophysis d) melanocytes e) parafollicular (C) cells

8. Which of the following is/are not mesodermal in origin?

- a) microglia b) bone and cartilage c) astrocytes
- d) cardiac muscle cells e) endothelium of blood and lymph vessels

9. Intermediate mesoderm gives rise to the

- a) neural tube b) heart c) kidneys and gonads d) somites e) notochord

10. During the third week of development, which statement is false?

- a) neural tube is formed.
- b) mesoderm exists between the ectoderm and endoderm everywhere.
- c) somites, which will give rise to the vertebrae, ribs and axial musculature, are formed.
- d) notochord forms the primitive axis of the embryo around which the axial skeleton will form.
- e) the primitive streak appears as a thickening of the embryonic epiblast at the caudal end of the embryonic disc.

11. During the fetal period,

- a) the spleen is the major site of erythropoiesis during the end of the 9th week.
- b) ossification of the fetal skeleton is active during the 13th -16th week.
- c) the bodies of 20-week fetuses are usually completely covered with lanugo hairs.
- d) by 18 weeks the uterus is formed and canalization of the vagina has begun.
- e) the amount of white fat is about 16% of body weight at 29 weeks.

12. A sacrocoxygealteratoma is a tumor that arises from remnants of the

- a) neural plate b) cloacal membrane c) posterior neuropore d) primitive streak e) notochord

13. Concerning the fetal membranes (developmental adnexa),

- a) the yolk sac helps control the embryo's body temperature.
- b) as the bladder enlarges, the umbilical cord involutes to form urachus.
- c) blood formation occurs in the wall of allantois during the third to fifth weeks.
- d) primordial germ cells appear in the endodermal lining of the wall of the allantois.

e) during the 4th week the endoderm of the yolk sac is incorporated into the embryo as the primordial gut.

14. Concerning multiple pregnancies,

a) the rate of dizygotic twinning increases with maternal age.

b) the frequency of monozygotic twinning shows marked racial differences.

c) twins with one amnion, one chorion and one placenta are always monozygotic.

d) dizygotic twins always have two amnions and two chorions. e) monozygotic twins are of the same sex.

15. Most birth defects occur during:

a) The first 2 weeks

b) The embryonic period

c) The fetal period

d) Early childhood

e) None of the above

16. The cranial half of the 1st cervical sclerotome gives rise to:

a) Upper part of body of the 1st cervical vertebra.

b) Lower part of body of the 1st cervical vertebra.

c) Whole body of the 1st cervical vertebra.

d) Part of base of occipital bone.

e) Part of roof of the cranial cavity.

17. Which of the following muscles develop from the epimeres?

a) Intercostal muscles.

b) The strap muscles of the neck.

c) Rectus abdominis.

d) Quadratus lumborum.

e) Extensor muscles of the back.

18. Failure of fusion of the neural arches of the developing vertebrae could lead to:

a) Achondroplasia

b) Acromegally

c) Spina bifida

d) Brachycephaly

e) Amelia

19. Which of the following gives rise to most muscles of the tongue?

a) Lateral mesoderm

b) Local splanchnic mesenchyme

c) Myotomes of the occipital somites

d) All of the above

e) None of the above

7. Frontonasal prominence gives rise to the:

a) Forehead

b) Sides of the face

c) Sides of the nose

d) All of the above

e) None of the above

8. Which of the following is not a component of vernix caseosa?

a) Secretions from sebaceous gland

b) Secretions of apocrine sweat glands

c) Fallen epidermal cells

d) Fallen hairs

e) None of the above

9. Structures derived from endoderm of the second pharyngeal pouch include:

a) Thymus gland

b) Palatine tonsil

c) Thyroid gland

d) Para-follicular cells of thyroid gland

e) All of the above

11. The muscular part of interventricular septum is formed by:

a) Downward extension of the spiral aorticopulmonary septum.

b) Fusion of superior and inferior endocardial cushions of the ventricular wall.

c) Fusion of right and left endocardial cushions of the ventricular wall.

d) Upward growth of a midline septum from the floor of a common ventricle.

e) Expansion of the lateral aspects of primitive ventricle with the midline region remained unexpanded.

83. Concerning the development of vasculature,

- a) distal parts of the 3rd pair of aortic arches join the dorsal aortae to form the internal carotid arteries.
- b) the left 4th aortic arch becomes the proximal part of the left subclavian artery.
- c) the distal part of the right 6th aortic arch becomes the ductus arteriosus.
- d) the superior vena cava forms from the right anterior cardinal vein and the right posterior cardinal vein.
- e) after birth, the proximal parts of the umbilical arteries persist as the superior vesical arteries.**

_____ 12. The common carotid arteries develop from:

- a) Truncus arteriosus
- b) 1st aortic arch arteries
- c) 2nd aortic arch arteries
- d) 3rd aortic arch arteries**
- e) Dorsal aortae

_____ 13. In fetal blood circulation oxygenated blood is not contaminated by deoxygenated blood in which of the following sites?

- a) Left atrium
- b) Right atrium
- c) Inferior vena cava
- d) Descending aorta
- e) None of the above

_____ 14. Which of the following veins participates in the formation of the inferior vena cava?

- a) Right vitelline vein**
- b) Right posterior cardinal vein
- c) Right umbilical vein
- d) Left umbilical vein
- e) Left posterior cardinal vein

_____ 15. Portal vein is derivative of:

- a) Right vitelline vein**
- b) Left vitelline vein
- c) Anastomosis between left and right vitelline veins
- d) Anastomosis between right umbilical and right vitelline veins
- e) Anastomosis between left umbilical and left vitelline veins

_____ 16. After birth umbilical vein becomes:

- a) Medial umbilical ligament**
- b) Median umbilical ligament
- c) Lateral umbilical ligament
- d) Ligamentum venosum
- e) Ligamentum teres hepatis**

_____ 17. Which of the following cells of the lung develop from mesoderm?

- a) Type I Pneumocytes.
- b) Type II Pneumocytes.
- c) Alveolar dust cells.**
- d) Clara cells
- e) Goblet cells

_____ 18. Concerning the **development of the GIT and associated structures**, which statement is **true**?

- a) Liver develops by fusion of ventral and dorsal hepatic buds.
- b) Stomach rotates 270° counterclockwise, similar to the midgut.
- c) The loop of the midgut has the superior mesenteric artery in its axis.**
- d) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.
- e) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion develops from the local mesoderm.

_____ 19. Which of the following statements is correct regarding to the **development of the gut**?

- a) Esophagus was short during early development, but gets elongated towards the end of embryonic period.

- b) The midgut was connected to the yolk sac by the allantois.
- c) The primitive gut tube is in open communication with the amniotic cavity.
- d) The site of junction of foregut and midgut derivatives is the 2nd part of duodenum just proximal to the entrance of common bile duct.
- e) All are correct

_____ 1. The **anterior 2/3rd of the tongue** develops at which pharyngeal arch level?

- a) 1st
- b) 2nd
- c) 3rd
- d) 4th
- e) 6th

_____ 2. Which of the following **salivary glands** develop from ectoderm?

- a) Parotid
- b) Submandibular
- c) Sublingual
- d) Lingual
- e) All of the above

_____ 19. Concerning the **development of pancreas**,

- a) The ventral bud arises from the proximal part of hepatic bud.

The ventral bud rotates & comes to lie anterior to the dorsal bud.

- c) The ventral bud arises from the distal part of foregut, while the dorsal bud arises from the proximal part of midgut.
- d) The exocrine portion develops from endoderm, while the endocrine portion develops from local mesoderm.
- e) The ventral bud appears before the appearance of the dorsal bud.

_____ 26. During the **development of the digestive system**, which statement is **true**?

- a) omphalocele is a persistence of the herniation of abdominal contents into the proximal part of the umbilical cord.
- b) the superior two-thirds of the adult anal canal are derived from proctodeum.
- c) the midgut may give rise to the descending colon and sigmoid colon.
- d) the dorsal pancreatic bud forms the uncinat process and part of the head of the pancreas.
- e) the anal membrane normally breaks down by the end of the second trimester.

_____ 34. The **midgut** begins with _____ and ends with _____.

- a) 1/2 jejunum; 2/3 transverse colon
- b) 1/2 ileum; 2/3 sigmoid colon
- c) 1/2 jejunum; 2/3 rectum
- d) 1/2 duodenum; 2/3 transverse colon
- e) 1/2 duodenum; 1/3 sigmoid colon

_____ 35. The **middle of the duodenum** would receive blood from:

- a) Superior mesenteric artery only
- b) Inferior mesenteric artery only
- c) Celiac artery only
- d) Celiac and inferior mesenteric artery
- e) Celiac and superior mesenteric artery

_____ 36. What two structures only form on the **ventral side of the bile duct**?

- a) Liver; Pancreas
- b) Liver; Gallbladder
- c) Pancreas; Gallbladder
- d) Heart; Liver
- e) Heart; Pancreas

_____ 37. During week six, the **midgut herniates** through the umbilical ring, rotating _____ degrees counterclockwise. Near week ten, the midgut returns, rotating _____ degrees for an overall rotation of _____ degrees.

- a) 180; 180; 0
- b) 180; 180; 360
- c) 90; 180; 90
- d) 90; 180; 270
- e) 270; 90; 180

_____ 38. A three-week old male infant comes into the Emergency Room for **weight loss and acute dehydration**. The infant has been **projectile vomiting** and the condition is worsening. Which of the following is most likely malformation?

- a) Gastric atresia
- b) Pyloric stenosis
- c) Annular pancreas
- d) Malrotation of the gut
- e) Intestinal duplication

_____ 39. **Meckel's diverticulum** is a congenital connection from the umbilicus via a vitelline ligament to

- a) Duodenum b) Jejunum c) Ileum d) Transverse colon e) Sigmoid colon

_____ **81.** With regard to the **development of the digestive system**,

- a) the foregut may give rise to the lower respiratory system.
- b) the midgut may give rise to the descending colon and sigmoid colon.
- c) the hindgut derivatives are supplied by inferior mesenteric artery.
- d) during the rotation of the stomach, the lesser curvature moves to the left and the greater curvature moves to the right.
- e) the dorsal pancreatic bud forms the uncinata process and part of the head of the pancreas.

_____ **82.** During the **development of the digestive system**,

- a) omphalocele is a persistence of the herniation of abdominal contents into the proximal part of the umbilical cord.
- b) the superior two-thirds of the adult anal canal are derived from proctodeum.
- c) the junction of the epithelium is derived from the ectoderm of the proctodeum.
- d) the inferior one-third of the anal canal is supplied with superior rectal artery.
- e) the anal membrane normally breaks down by the end of the second trimester.

_____ **20.** Which of the following statements is **not correct** regarding to the **development of the coelomic cavities?**

- a) Peritoneal cavity was temporarily divided into two by the gut and its primitive mesenteries.
- b) Develop within the intermediate mesoderm.
- c) Are lined by both splanchnic and somatic mesoderm layers.
- d) Primitive pericardial cavity gives rise to the serous pericardium.
- e) Pericardioperitoneal canals are invaginated by the growing lungs.

_____ **21.** The **crura of the diaphragm** develops from:

- a) Septum transversum.
- b) Dorsal mesentery of the esophagus.
- c) Pleuroperitoneal membrane.
- d) Myotomes of the body wall.
- e) None of the above.

_____ **25.** Concerning the development of the **Integumentary system**, which statement is **true?**

- a) keratinocytes from the epidermis mix with secretions of sebaceous glands to form the vernix caseosa.
- b) toenails and fingernails begin to develop at the tips of the digits postnatally.
- c) teeth develop from ectoderm, mesoderm and neural crest cells.
- d) eccrine sweat glands develop from mesenchyme, which is derived from the mesoderm underlying the surface ectoderm.
- e) the epidermis is derived from surface ectoderm.

_____ **3.** All of the following ***appendages of the skin*** develop from ectoderm, **except:**

a) Hairs. b) Sweat glands. c) Sebaceous glands. d) Arrector pili muscle. e) Nails.

28. What germ layer is the **dermis** derived from?

a) Paraxial mesoderm b) Intermediate mesoderm c) Lateral plate mesoderm
d) Neural crest e) Endoderm

29. At birth the infant is covered by a whitish paste (vernix caseosa), which protects the skin against the macerating action of amniotic fluid. Which of the following is the **vernix caseosa** derived from?

a) Arrector pili muscles b) Dermal root sheaths c) Hair papilla
d) Sebaceous glands e) Hair blood vessels

30. During a routine examination of a female patient, **accessory nipples** are found in the axilla and on the mid thorax along the embryological mammary ridge. What is the terminal extension of this ridge?

a) Mid thorax b) Diaphragm c) Umbilicus d) Pubic region e) Lower thigh to knee

31. From what layer of the epidermis do **hair follicles** begin to proliferate?

a) Stratum corneum b) Stratum lucidum c) Stratum granulosum
d) Stratum spinosum e) Stratum germinativum

32. This disorder results from **excessive keratinization of the skin**, whereby the skin appears as very dry and scaly. In extreme cases (harlequin fetus), the skin is abnormally thick and cracked.

a) Albinism b) Amelogenesis imperfecta c) Leprosy d) Psoriasis e) Ichthyosis

33. A patient presents with **red blotches on their face and neck**. They say they've had these birthmarks as long as they can remember and they were told it has something to do with capillaries. Which of the following malformation does this patient likely have?

a) Albinism b) Angioma c) Ichthyosis d) Collodion e) Thalidomide defect

5. With regard to the **congenital anomalies of the kidneys and ureters** which statement is **false**?

a) Accessory renal arteries are end arteries.
b) Bilateral renal agenesis is associated with oligohydramnios.
c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.
d) Unilateral renal agenesis is more common in males than females. e) None of the above.

6. Which of the following is not the **adult derivative of the ureteric bud**?

a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

7. **Urachal cyst** is a remnant of:

a) urogenital sinus b) urogenital ridge c) cloaca d) allantois e) mesonephric duct

8. The **podocytes of Bowman's capsule** are derived from:

a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

40. Concerning the **development of the urinary system**, which statement is **false**?

a) Urinary system begins to develop before the genital system
b) The metanephric diverticulum is the primordium of the ureter.
c) Bilateral renal agenesis is associated with oligohydramnios.

- d) Differentiation of the nephrons depends upon induction by the collecting tubules.
 e) The kidneys attain their definitive adult position toward the end of the fetal period.

_____ **18.** Which of the following is not the adult derivative of the **ureteric bud**?

- a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

_____ **19.** Concerning the **development of the urinary bladder and urethra**, which statement is **false**?

- a) The female urethra is derived from endoderm of the urogenital sinus.
 b) In adults, the urachus is represented by the median umbilical ligament.
c) The urinary bladder becomes a pelvic organ during the 9th week of development.
 d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
 e) Epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

_____ **11.** During the **development of the kidney systems**, which statement is **true**?

- a) The transitional epithelium lining the urethra is derived from neuroectoderm.
b) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
 c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.
 d) Urine production begins shortly after birth. e) At birth, the kidneys have a lobulated appearance.

_____ **41.** Concerning the **development of the urinary bladder and urethra**, which statement is **false**?

- a) The female urethra is derived from endoderm of the urogenital sinus.
 b) In adults, the urachus is represented by the median umbilical ligament.
c) The urinary bladder becomes a pelvic organ during the 9th week of development.
 d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
 e) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

_____ **40.** During the **pronephros period** (week 3), the first kidney develops in the _____ region.
 By week 4 this kidney degenerates.

- a) Cervical b) Thoracic c) Lumbar d) Sacral e) Cloaca

_____ **41.** The **ureteric bud** gives rise to the _____ and the **metanephric mesoderm** gives rise to the _____.

- a) Nephrons; Collecting system b) Collecting system; Nephrons c) Bowman's capsule; Ureter
 d) Loop of Henle; Renal pelvis e) Distal/proximal convoluted tubules; Major/minor calyces

_____ **42.** The **urorectal septum** divides the _____ from the _____.

- a) Anorectal canal; Urogenital sinus b) Urogenital sinus; Mesonephric duct c) Urachus; Allantois
 d) Urinary bladder; Allantois e) Mesonephric duct; Ureteric bud

_____ **43.** The **urachus** forms the:

- a) Allantois b) Median umbilical ligament c) Medial umbilical ligament
 d) Lateral umbilical ligament e) Round (umbilical) ligament

_____ **44.** The **bladder** becomes continuous with the lower end of the mesonephric (Wolffian) duct, which will form the _____ of the bladder.

- a) Urethra b) Prostate c) Trigone d) Posterior wall e) Anterior wall

_____ **45.** At birth, a child has **large palpable flank masses and abnormal extremities**. The delivery physician is very concerned as she knows this **autosomally-dominant disorder** can lead to renal failure and the child will likely need a kidney transplant to survive. Which of the following disorders does the child likely have?
a) Polycystic kidney _____ b) Renal agenesis c) Pelvic kidney d) Bifid ureter e) Bifid kidney

_____ **46.** This disorder, associated with **oligohydramnios**, occurs when the ureteric bud fails to reach the metanephric mesoderm.

- a) Polycystic kidney _____ b) Renal agenesis c) Pelvic kidney
d) Horseshoe kidney e) Bifid ureter

_____ **47.** What artery prevents the **horseshoe kidney** (metanephric mesodermal fusion) from ascending high into the abdomen?

- a) Celiac trunk b) Superior mesenteric artery c) Inferior mesenteric artery
d) Left renal artery e) Right renal artery

_____ **48.** A newborn baby presents with fluid draining from the **umbilicus onto the skin**. Testing of the fluid identifies it as urine. What is the most likely diagnosis?

- a) Urachal cyst b) Urachal fistula c) Exstrophy of the bladder d) Bifid ureter e) Horseshoe kidney

_____ **43.** With regard to the **development of the endocrine system**, which of the following statements is **false**?

- a) the pituitary gland is ectodermal in origin.
b) the thyroid gland is the first endocrine gland to develop in the embryo.
c) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.
d) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
e) the oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.

_____ **9.** With regard to the development of the **endocrine system**, which of the following statements is **false**?

- a) the oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.
b) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
c) the thyroid gland is the first endocrine gland to develop in the embryo.
d) the pituitary gland is ectodermal in origin.
e) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.

_____ **44.** With regard to the **development of gonads**, which statement is **false**?

- a) The surface epithelium of the female gonad develop into oogonia.
b) Sex of an embryo is determined genetically at the time of fertilization.
c) Germ cells do not appear in the genital ridges until the 6th week of development.
d) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ **45.** With regard to the **vestigial structures derived from embryonic genital system**, which statement is **false**?

- a) A structure found within the adult female pelvis formed from the gubernaculum is median umbilical ligament

- b) The cranial end of mesonephric duct in males may persist as paradidymis.
- c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis.
- d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

46. During the development of the female reproductive system, which statement is false?

- a) The phallus forms the mons pubis.
- b) The labioscrotal swellings form the labia majora.
- c) The urogenital folds remain unfused and form the labia minora.
- d) The cranial portion of the paramesonephric ducts form the uterine tubes.
- e) During development of the ovaries, the primitive sex (medullary) cords degenerate.

47. During the development of the male reproductive system, which statement is false?

- a) Hypospadias is constantly associated with exstrophy of urinary bladder.
- b) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
- c) Sertoli cells, derived from mesoderm, constitute most of the seminiferous epithelium in the fetal testis.
- d) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.
- e) The epididymis, ductus deferens, ejaculatory duct, and seminal vesicle all arise from the paramesonephric ducts.

10. A structure found within the adult female pelvis formed from the gubernaculum is

- a) broad ligament
- b) superior ligament of the ovary
- c) medial umbilical ligament
- d) round ligament of the uterus
- e) median umbilical ligament

48. During the development of the reproductive system, which statement is true?

- a) The lower portion of the vagina is derived from hymen.
- b) Gubernaculum does play a pivotal role in the descent of ovaries.
- c) Dihydrotestosterone inhibits the growth of the paramesonephric duct in males.
- d) In the male, failure of urethral folds to fuse completely results in cryptorchidism.
- e) The most common cause of testicular feminization [Androgen Insensitivity] syndrome is inadequate production of testosterone and müllerian-inhibiting factor.

12. The labia minora arise embryologically from:

- a) phallus
- b) labioscrotal swellings
- c) sinovaginal bulbs
- d) urogenital folds
- e) paramesonephric duct

13. The uterine tubes of the adult female are derived embryologically from

- a) mesonephric duct
- b) mesonephric tubules
- c) paramesonephric duct
- d) paramesonephric tubules
- e) uterovaginal primordium

14. The most common cause of testicular feminization [Androgen Insensitivity] syndrome is

- a) a 46, XO genotype
- b) a 47, XXY genotype
- c) lack of androgen receptors
- d) congenital adrenal hyperplasia
- e) inadequate production of testosterone and müllerian-inhibiting factor.

15. In the male, failure of urethral folds to fuse completely results in

- a) hypospadias
- b) epispadias
- c) cryptorchidism
- d) hydrocele
- e) congenital inguinal hernia

21. During the development of the female reproductive system, which statement is false?

- a) The urogenital folds remain unfused and form the labia minora.
- b) The cranial portion of the paramesonephric ducts form the uterine tubes.
- c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.
- d) The phallus forms the mons pubis.
- e) The labioscrotal swellings form the labia majora.

_____ 22. Concerning the **development of the genital system**, one statement is **false**?

- a) Bilateral cryptorchidism may result in sterility.
- b) Hypospadias is constantly associated with exstrophy of urinary bladder.
- c) In testicular feminization syndrome, uterus and uterine tubes are generally absent.
- d) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora
- e) In epispadias usually the penis is underdeveloped and is curved ventrally, a condition known as chordee.

_____ 23. With regard to the **development of gonads**, which statement is **false**?

- a) The surface epithelium of the female gonad develop into oogonia.
- b) Sex of an embryo is determined genetically at the time of fertilization.
- c) Germ cells do not appear in the genital ridges until the 6th week of development.
- d) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
- e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ 24. With regard to the **vestigial structures derived from embryonic genital ducts**, which statement is **false**?

- a) The prostatic utericle is homologous to the vagina.
- b) The cranial end of mesonephric duct in males may persist as paradidymis.
- c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis
- d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ 49. A patient is born with karyotype 46, XY but has no problem with production of testosterone or müllerian inhibiting hormone. Their **vagina ends in a blind pouch and testes are lodged in the inguinal canal**. Which of the following is the best description of this patient:

- a) Male pseudo-hermaphrodite
- b) Female pseudo-hermaphrodite
- c) Androgen insensitivity syndrome
- d) Turner's syndrome
- e) Klinefelter syndrome

_____ 50. During development of the **ovaries, the primitive sex (medullary) cords** _____.

- a) Proliferate
- b) Move laterally
- c) Move medially
- d) Differentiate
- e) Degenerate

_____ 51. The **gubernaculum** is attached to the uterus and as the ovaries descend it becomes the _____ of the uterus.

- a) Canal of Nuck
- b) Ovarian ligament
- c) Round ligament
- d) Ovarian and round ligaments
- e) Gubernacular ligament

_____ 52. **Development of the testis** is induced by the sry-gene, which encodes testis-determining factor (TDF). TDF causes all of the following **EXCEPT**:

- a) The sex cords continue to proliferate to form the testis.
- b) Leydig cells are secreting testosterone thus have very little SER.
- c) Towards the hilum, the cords grow into a network that will give rise to the rete testes.
- d) During further development, a thick layer (tunica albuginea) separates the cords from the surface epithelium.

e) Sex cords become seminiferous cords that form seminiferous tubules (Sertoli cells), straight tubules, and rete testes

_____ **53.** Which of the following inhibits the growth of the **paramesonephric duct (males)**?

- a) Testosterone b) Leydig Cells c) Dihydrotestosterone d) Estrogen e) Müllerian substances (MIFs)

_____ **54.** As the **testes descend**, they follow the:

- a) Processus vaginalis b) Parietal layer of tunica vaginalis c) Visceral layer of tunica vaginalis
d) Caudal genal ligament e) Cryptorchidism

_____ **55.** The **epididymis, ductus deferens, ejaculatory duct, and seminal vesicle** all arise from:

- a) The Müllerian ducts b) The paramesonephric ducts c) The mesonephric ducts
d) Leydig hormones e) Wolffian hormones

_____ **56.** The **uterine tubes, uterus, cervix, and superior part (1/3) of the vagina** all arise from:

- a) The Wolffian ducts b) The paramesonephric ducts c) The mesonephric ducts
d) Leydig hormones e) Müllerian hormones

_____ **57.** The **epioophoron and paroophoron**, which are important as cysts, arise from:

- a) The Müllerian ducts b) The paramesonephric ducts c) The mesonephric ducts
d) Leydig hormones e) Wolffian hormones

_____ **58.** The **lower portion of the vagina** is derived from:

- a) Sinovaginal bulb b) Uteral lumen c) Hymen d) Urogenital sinus e) Vaginal plate

_____ **59.** What structure in the **indifferent stage** (weeks 3-6) becomes the **clitoris**?

- a) Urogenital groove b) Genital swelling c) Urethral fold d) Perineum e) Genital tubercle

_____ **60.** **Genetic 46,XY** would be:

- a) Turner's syndrome b) Klinefelter syndrome c) True hermaphrodite
d) Male pseudo-hermaphrodite e) Female pseudo-hermaphrodite

_____ **23.** Which of the following nuclei is **not** developed from the **alar plate of the neural tube**?

- a) Olivary nucleus b) Cuneate nucleus c) Thalamus
d) Superior colliculus e) None of the above

_____ **24.** **Primary brain vesicle** that remains relatively unchanged as compared to the others is:

- a) Rhombencephalon b) Prosencephalon c) Telencephalon
d) Mesencephalon e) Diencephalons

_____ **25.** During the **development of the central nervous system** zone of the wall of the neural tube to be occupied by neuroblasts forming the gray matter is:

- a) Marginal zone b) Ventricular zone c) Mantle zone d) Ependymal zone e) Neuroepithelial zone

_____ **49.** With regard to **development of the eye**, which statement is **true**?

- a. The hyaloid canal is found in the vitreous body.
- b. Aqueous humor is produced by the choroid plexus.
- c. Failure of choroid fissure to close results in congenital aphakia.
- d. The optic nerve is a tract of diencephalon that is completely myelinated during the third trimester.
- e. Abnormal development of canal of Schlemm or the iridocorneal filtration angle causes congenital cataract.

_____ **50.** With regard to **development of the ear**, which statement is **true**?

- a. The spiral organ of Corti is derived from neural crest.
- b. The middle ear cavity develops from the otic vesicle.
- c. The utricular portion of the otic vesicle gives rise to cochlear duct.
- d. Perilymph enters the subarachnoid space via the cochlear duct.
- e. None of the above.

_____ **1.** Most **birth defects** occur during which of the following periods/stages of the intrauterine life?

- a) First 2 weeks b) Embryonic period c) Fetal period d) Last trimester e) Zygote

_____ **2.** Which of the following statements is **not true** regarding the **human placenta**?

- a) Maternal portion is formed by decidua parietalis. b) Fetal portion is formed by chorion frondosum.
- c) Cotyledons are seen from the maternal aspect. d) Fetal portion is smooth.
- e) At full term covers the whole surface area of the endometrium.

_____ **3.** The central tendon of the diaphragm develops from:

- a) Dorsal mesentery of the esophagus b) Pleuroperitoneal membranes
- c) Pleuropericardial membranes. d) Lateral body wall e) Septum transversum.

_____ **4.** Which of the following is not a component of **vernix caseosa**?

- a) Secretions from sebaceous gland b) Secretions of apocrine sweat glands
- c) Fallen epidermal cells d) Fallen hairs e) None of the above

_____ **5.** Structure that is **not** developed in the **third branchial arch** include:

- a) Lesser cornu of hyoid bone b) Body of the hyoid bone
- c) Glossopharyngeal nerve d) Common carotid artery e) Stylopharyngeus muscle

_____ 6. Which of the following structures develop from **endoderm of the second pharyngeal pouch**?

- a) Thymus gland b) Palatine tonsil c) Thyroid gland
d) Superior parathyroid gland e) Parafollicular cells of thyroid gland

_____ 7. The **anterior 2/3rd of the tongue** develops at which pharyngeal arch level?

- a) 1st b) 2nd c) 3rd d) 4th e) 6th

_____ 8. Which of the following **salivary glands** develop from ectoderm?

- a) Parotid b) Submandibular c) Sublingual d) Lingual e) All of the above

_____ 9. **Secretion of surfactant, which is an important substance for survival of premature infants appears during which of the following periods of lung development?**

- a) Canalicular. b) Bronchiolar. c) Terminal sac period d) Pseudoglandular e) Alveolar

_____ 10. Regarding the **development of the gut**, which of the following statements is **correct**?

- a) The lumen of the intestine is temporarily blocked by epithelial proliferation.
b) The site of junction of foregut and midgut derivatives is the middle of jejunum.
c) Is formed as the result of absorption of part of the amniotic cavity because of the embryonic foldings.
d) The midgut is temporarily connected to the yolk sac by the allantois.
e) All of the above.

_____ 11. All of the following **appendages of the skin** develop from ectoderm, except:

- a) Hairs. b) Sweat glands. c) Sebaceous glands. d) Arrector pili muscle. e) Nails.

_____ 12. **With regard to fertilization, which statement is false?**

- a) restores the normal diploid number of chromosomes in the zygote.
b) results in variation of human species through mingling of maternal and paternal chromosomes.
c) causes metabolic activation of the ovum and initiates cleavage of the zygote.
d) determines chromosomal sex of the embryo. e) None of the above.

_____ 13. During the **first week of development**,

- a) the embryo is a solid ball of blastomeres and is called gastrula.
b) mesoderm is formed which will give rise later to the placental membranes.
c) the blastocyst is superficially implanted in the uterine wall (endometrium).
d) developmental disturbances may give rise to major congenital anomalies.
e) the amniotic cavity is developed within the yolk sac as a space between the cytotrophoblast and the embryoblast.

_____ 14. During the **third week of development**, which statement is **false**?

- a) somites, which will give rise to the vertebrae, ribs and axial musculature, are formed.
b) notochord forms the primitive axis of the embryo around which the axial skeleton will form.
c) the primitive streak appears as a thickening of the embryonic epiblast at the caudal end of the embryonic disc.
d) mesoderm exists between the ectoderm and endoderm everywhere. e) neural tube is formed.

_____ 15. During the **fetal period**,

- a) the spleen is the major site of erythropoiesis during the end of the 9th week.
b) ossification of the fetal skeleton is active during the 13th -16th week.
c) the bodies of 20-week fetuses are usually completely covered with lanugo hairs.

- d) by 18 weeks the uterus is formed and canalization of the vagina has begun.
- e) the amount of white fat is about 16% of body weight at 29 weeks.

16. With regard to the placenta,

- a) the maternal part of the placenta develops from the chorionic sac.
- b) the fetal part of the placenta is attached to the maternal part of the placenta by the cytotrophoblastic shell.
- c) amniochorionic membrane ruptures during labor.
- d) as pregnancy advances, the placental membrane becomes progressively thinner.
- e) the fetal surface of a freshly delivered placenta has the characteristic cobblestone appearance produced by cotyledons.

17. Concerning the fetal membranes (developmental adnexa),

- a) the yolk sac helps control the embryo's body temperature.
- b) primordial germ cells appear in the endodermal lining of the wall of the allantois.
- c) during the 4th week the endoderm of the yolk sac is incorporated into the embryo as the primordial gut.
- d) as the bladder enlarges, the umbilical cord involutes to form urachus.
- e) blood formation occurs in the wall of allantois during the third to fifth weeks.

18. Concerning multiple pregnancies,

- a) twins with one amnion, one chorion and one placenta are always monozygotic.
- b) the rate of dizygotic twinning increases with maternal age.
- c) the frequency of monozygotic twinning shows marked racial differences.
- d) dizygotic twins always have two amnions and two chorions.
- e) monozygotic twins are of the same sex.

19. The neuroepithelial cells of the neural tube gives rise to all of the following cells of the brain except:

- a) Microglia.
- b) Astrocytes.
- c) Oligodendrocytes.
- d) Purkinje cells of cerebellar cortex.
- e) Ependymal cells.

20. With regard to birth defects,

- a) the phenotype of Turner syndrome is female.
- b) thalidomide induces limb defects.
- c) trisomy of the autosomes occurs with increasing frequency as maternal age increases.
- d) deafness has been reported in infants of mothers who have been treated with penicillin during the 2nd trimester.
- e) phenobarbital is considered to be a safe, antiepileptic drug for use during pregnancy.

21. Concerning the development of the GIT and associated structures:

- a) The loop of the midgut has the superior mesenteric artery in its axis.
- b) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion develops from the local mesoderm.
- c) Liver develops by fusion of ventral and dorsal hepatic buds.
- d) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.
- e) Stomach rotates 270° counterclockwise, similar to the midgut.

22. With regard to the development of the musculoskeletal system,

- a) the muscular system develops from neuroectoderm.
- b) the upper limbs rotate laterally through 90° on their longitudinal axes.
- c) premature closure of the frontal (metopic) suture results in scaphocephaly.
- d) sclerotomes of the somites form the viscerocranium.
- e) xiphoid process develops during the 6th week.

_____ **23. Cerebellum** develops from cells of:

- a) Basal plate of myelencephalon
- b) Alar plate of myelencephalon
- c) Basal plate of metencephalon
- d) Alar plate of myelencephalon
- e) Basal plate of mesencephalon

_____ **24. Primary brain vesicle** that remains relatively unchanged into the secondary brain vesicle is:

- a) Rhombencephalon
- b) Prosencephalon
- c) Telencephalon
- d) Mesencephalon
- e) Diencephalon

_____ **25. Which of the following components plays the most active role in invading the endometrium during blastocyst implantation?**

- a) epiblast
- b) syncytiotrophoblast
- c) hydroblast
- d) extraembryonic somatic mesoderm
- e) extraembryonic visceral mesoderm

_____ **26. The prochordal plate** marks the site of the future

- a) umbilical chord
- b) heart
- c) mouth
- d) anus
- e) nose

_____ **27. Which process establishes the three definitive germ layers?**

- a) neurulation
- b) gastrulation
- c) craniocaudal folding
- d) lateral folding
- e) angiogenesis

_____ **28. The spiral organ of Corti** is derived from

- a) both ectoderm and mesoderm
- b) neural crest
- c) endoderm
- d) mesoderm
- e) surface ectoderm

_____ **29. The middle ear cavity**

- a) is of mesodermal origin
- b) develops from the otic vesicle
- c) develops from pharyngeal arch 1
- d) develops from pharyngeal pouch 2
- e) develops from pharyngeal pouch 1

_____ **29. With regard to the congenital anomalies of the kidneys and ureters which statement is false?**

- a) Accessory renal arteries are end arteries.
- b) Bilateral renal agenesis is associated with oligohydramnios.
- c) Incontinence is the common complaint resulting from an ectopic ureteric orifice in both sexes.
- d) Unilateral renal agenesis is more common in males than females.
- e) None of the above.

_____ **30. The saccular portion of the otic vesicle** gives rise to the

- a) organ of Corti
- b) endolymphatic duct
- c) superior semicircular canal
- d) crus commune nonampullare
- e) lateral semicircular canal

_____ **31. Perilymph** enters the subarachnoid space via the

- a) cochlear duct
- b) ductus reuniens
- c) perilymphatic duct
- d) vestibular aqueduct
- e) utriculosaccular duct

_____ **32. Which of the following structures in the eye** is derived from **surface ectoderm**?

- a) retina
- b) dilator and sphincter pupillae muscles of iris
- c) corneal endothelium
- d) vitreous body
- e) lens

_____ **33. Which of the following is not the adult derivative of the ureteric bud?**

- a) renal pelvis
- b) major calyx
- c) minor calyx
- d) collecting duct
- e) connecting tubule

_____ **34. Urachal cyst** is a remnant of

- a) urogenital sinus b) urogenital ridge c) cloaca d) allantois e) mesonephric duct

_____ **35. The podocytes of Bowman's capsule** are derived from

- a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

_____ **36. A structure found within the adult female pelvis formed from the gubernaculum** is

- a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament

_____ **37. After the sinovaginal bulbs** have proliferated and fused, they form a solid core of endodermal cells called

- a) vestibule of the vagina b) uterovaginal primordium c) urogenital sinus d) vaginal plate
e) clitoris

_____ **38. The labia minora** arise embryologically from

- a) phallus b) labioscrotal swellings c) sinovaginal bulbs
d) urogenital folds e) paramesonephric duct

_____ **39. The uterine tubes** of the adult female are derived embryologically from

- a) mesonephric duct b) mesonephric tubules c) paramesonephric duct
d) paramesonephric tubules e) uterovaginal primordium

_____ **40. The most common cause of testicular feminization syndrome is**

- a) a 46, XO genotype b) a 47, XXY genotype
c) lack of androgen receptors d) congenital adrenal hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

_____ **41. In the male, failure of urethral folds to fuse completely** results in

- a) hypospadias b) epispadias c) cryptorchidism d) hydrocele e) congenital inguinal hernia

_____ **42. The Y chromosome** carries a gene on its short arm that codes for

- a) testosterone b) MIF c) testes-determining factor (TDF) d) progesterone e) estrogen

_____ **48. Which of the following best describes the placental components of dizygotic twins?**

- a) one placenta, two amniotic sacs, one chorion b) one placenta, two amniotic sacs, two chorions
c) two placentas, two amniotic sacs, one chorion d) two placentas, two amniotic sacs, two chorions
e) one placenta, two amniotic sacs, two chorions

_____ **43. The notochord** induces the overlying ectoderm to develop into the

- a) skin b) eye c) primitive streak d) primitive node e) neural plate

_____ **44. A sacroccygeal teratoma** is a tumor that arises from remnants of the

- a) neural plate b) cloacal membrane c) posterior neuropore d) primitive streak e) notochord

_____ **45. Intermediate mesoderm** gives rise to the

- a) neural tube b) heart c) kidneys and gonads d) somites e) notochord

- _____ **46.** The **proximal part of the right subclavian artery** is derived from aortic arch
 a) 1 b) 2 c) 3 d) 4 e) 6
- _____ **47.** The **portal vein** is derived from
 a) vitelline veins b) umbilical veins c) anterior cardinal veins
 e) posterior cardinal veins d) subcardinal veins
- _____ **49.** The **stapedius muscle** that moves the stapes ossicle is innervated by
 a) CN V3 b) CN XII c) CN III d) CN VII e) cervical nerves C2 and C 3
- _____ **50.** **Pharyngeal groove 1** gives rise to
 a) internal auditory meatus b) external auditory meatus
 c) eustachian tube d) cervical sinus e) primary tympanic cavity
- _____ **51.** **Failure of choroid fissure to close** results in
 a) congenital detached retina b) congenital aniridia c) congenital aphakia
 d) colobomairidis e) microphthalmos
- _____ **52.** **Aqueous humor** is produced by the
 a) choroid plexus b) trabecular meshwork c) ciliary processes
 d) vitreous body e) lens vesicle
- _____ **53.** Concerning the **development of the urogenital system**, which of the following statements is true?
 a) The labia minora arise embryologically from paramesonephric ducts.
 b) The uterine tubes of the adult female are derived embryologically from uterovaginalprimordium.
 c) The most common cause of testicular feminization syndrome is the presence of abundant androgen receptors.
 d) Round ligament of the uterus is formed from the gubernaculums.
 e) The X chromosome carries a gene on its short arm that codes for TDF.
- _____ **54.** With regard to the development of the **endocrine system**, which of the following statements is false?
 a) theoxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.
 b) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
 c) the thyroid gland is the first endocrine gland to develop in the embryo.
 d) the pituitary gland is ectodermal in origin.
 e) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.
- _____ **55.** The **transitional epithelium lining the ureter** is derived from
 a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells
- _____ **56.** Which of the following is not the adult derivative of the **ureteric bud**?
 a) renal pelvis b) major calyx c) minor calyx d)collecting duct e) connecting tubule
- _____ **57.** Which of the following statements about the **optic nerve** is false? The **optic nerve**

- a. is a tract of diencephalon. b. is myelinated by oligodendrocyte. c. is invested by meninges.
d. is not capable of regeneration after transaction. e. is developed from the first pharyngeal pouch.

_____ **58. Concerning the development of the Urinary System, which statement is false?**

- a) Urinary system begins to develop before the genital system
- b) The metanephric diverticulum is the primordium of the ureter.
- c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- d) Differentiation of the nephrons depends upon induction by the collecting tubules.
- e) The kidneys attain their definitive adult position toward the end of the fetal period.

_____ **60. Concerning the development of the urinary bladder and urethra, which statement is false?**

- a) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- b) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.
- c) In adults, the urachus is represented by the median umbilical ligament.
- d) The urinary bladder becomes a pelvic organ during the 9th week of development.
- e) The female urethra is derived from endoderm of the urogenital sinus.

_____ **63. During the development of the kidney systems, which statement is true?**

- a) The transitional epithelium lining the urethra is derived from neuroectoderm.
- b) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
- c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.
- d) Urine production begins shortly after birth. e) At birth, the kidneys have a lobulated appearance.

_____ **64. During the development of the female reproductive system, which statement is false?**

- a) The urogenital folds remain unfused and form the labia minora.
- b) The cranial portion of the paramesonephric ducts form the uterine tubes.
- c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.
- d) The phallus forms the mons pubis. e) The labioscrotal swellings form the labia majora.

_____ **65. With regard to the development of the male reproductive system, which statement is false?**

a) The genotype of an embryo is established at fertilization.

b) By week 12, male characteristics of the external genitalia are recognizable.

c) Sertoli cells are derived from mesoderm.

d) Seminiferous tubules are formed by week 20.

e) In epispadias usually the penis is underdeveloped and curved ventrally a condition known as chordee.

_____ **66. Concerning the development of the genital system, one statement is false?**

a) Hypospadias is constantly associated with exstrophy of urinary bladder.

b) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora.

c) In testicular feminization syndrome, uterus and uterine tubes are generally absent.

d) Bilateral cryptorchidism results in sterility.

e) None of the above.

_____ **68. With regard to the development of gonads, which statement is false?**

a) Sex of an embryo is determined genetically at the time of fertilization.

b) Germ cells do not appear in the genital ridges until the 6th week of development.

c) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.

d) The surface epithelium of the female gonad develop into oogonia.

e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ **71. During the development of the testes, which statement is false?**

a) A dense tunica albuginea is the characteristic feature of testicular development in the fetus.

b) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.

c) The fetal testis produces antimüllerian hormone by sertoli cells at 6-7 weeks

d) Sertoli cells are derived from the surface epithelium of the testis.

e) None of the above.

_____ **69. With regard to the vestigial structures derived from embryonic genital ducts, which statement is false?**

a) The cranial end of mesonephric duct in males may persist as paradidymis.

b) The epoophoron in the females correspond to the efferent ductules and duct of epididymis

- c) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- d) The prostatic utricle is homologous to the vagina.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ **70. Concerning the development of the genital system, which statement is false?**

- a) Inguinal canals develop in both sexes.
- b) Sertoli cells constitute most of the seminiferous epithelium in the fetal testis.
- c) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
- d) Gubernaculum does not play any role in the descent of ovaries.
- e) The persistent processus vaginalis in the female is called canal of nuck.

_____ **72. With regard to development of the ovaries, which statement is false?**

- a) The cranial part of the gubernaculum becomes the ovarian ligament.
- b) As the ovary separates from the regressing mesonephros, it is suspended by mesovarium.
- c) The surface epithelium becomes separated from the follicles in the cortex by tunica albuginea.
- d) The primordial follicles are derived from the rete ovarii.
- e) No oogonia are formed postnatally.

_____ **73. With regard to development of the eye, which statement is true?**

- a. Failure of choroid fissure to close results in congenital aphakia.
- b. The epithelium of the ciliary body is derived from endoderm.
- c. The optic nerve is a tract of diencephalon that is completely myelinated during the third trimester.
- d. The hyaloid canal (cloquet's canal) is found in the vitreous body.
- e. Aqueous humor is produced by the choroid plexus.

_____ **74. With regard to development of the ear, which statement is true?**

- a. The spiral organ of Corti is derived from neural crest.
- b. The middle ear cavity develops from the otic vesicle.

c. The utricular portion of the otic vesicle gives rise to cochlear duct.

d. Perilymph enters the subarachnoid space via the endolymphatic sac.

e. None of the above.

_____ **76. Abnormal development of canal of Schlemm or the iridocorneal filtration angle causes**

a. colobomairidis

b. congenital cataract

c. congenital glaucoma

d. cyclopia

e. microphthalmia

_____ **78. With regard to the development of the head and neck,**

a) sphenoidal sinuses develop during the early fetal life (10 to 11 weeks)

b) the critical period of palate development is from the end of the 6th week until the beginning of the 9th week.

c) taste buds form on the dorsal surface of the tongue only.

d) all parenchymal (secretory) tissue of the salivary glands arises by proliferation of the oral epithelium.

e) by late fetal period, the nasolacrimal duct drains into the inferior meatus in the lateral wall of the nasal cavity.

_____ **79. Concerning the development of vasculature,**

a) distal parts of the 3rd pair of aortic arches join the dorsal aortae to form the internal carotid arteries.

b) the left 4th aortic arch becomes the proximal part of the left subclavian artery.

c) the distal part of the right 6th aortic arch becomes the ductus arteriosus.

d) the superior vena cava forms from the right anterior cardinal vein and the right posterior cardinal vein.

e) after birth, the proximal parts of the umbilical arteries persist as the superior vesical arteries.

_____ **77. Utricular portion of the otic vesicle does not give rise to**

a. utricle

b. semicircular ducts

c. vestibular ganglion of CN VII

d. endolymphatic duct and sac

e. cochlear duct [organ of Corti]

_____ **80. Development of the Integumentary system entails that,**

a) the epidermis is derived from surface ectoderm.

b) castoff cells from the epidermis mix with secretions of sebaceous glands to form the vernix caseosa.

c) teeth develop from ectoderm, mesoderm and neural crest cells.

d) eccrine sweat glands develop from mesenchyme, which is derived from the mesoderm underlying the surface ectoderm.

e) toenails and fingernails begin to develop at the tips of the digits postnatally.

GOOD LUCK